AN ETALE REPRESENTABLE MORPHISM OF FORMAL ALGEBRAIC SPACES INDUCES A COCONTINUOUS FUNCTOR ON THE SMALL ETALE SITES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The corresponding statement for algebraic spaces. Properties of Spaces, Lemma spaces-properties-lemma-etale-morphism-topoi, invoked at proof lines 7464-7466. Provides the assertion once the induced morphism of reductions is known to be etale.
- An algebraic space etale over a reduced algebraic space is reduced, and the reduction of the source is the base change of the reduction of the target along the morphism. Properties of Spaces, Section spaces-properties-section-types-properties at proof lines 7468-7473, and Same chapter, Lemma lemma-reduction-smooth at proof lines 7473-7475. Identifies the reduction of the source with the base change and gives its reducedness.

Source reference. Original lemma 00XO: sources/stacks-chow/tags.txt, source lines 1226–1226. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00XR: sources/stacks-chow/tags.txt, source lines 1229–1229. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03LP: sources/stacks-chow/tags.txt, source lines 4721–4721. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 03E5: sources/stacks-chow/tags.txt, source lines 4444–4444. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Let S be a scheme, and let $f : X \to Y$ be a morphism of formal algebraic spaces over S. Assume f is representable by algebraic spaces and étale. In this case there is a cocontinuous functor $j : X_{étale} \to Y_{étale}$. The morphism of topoi f_{small} is the morphism of topoi associated to j, see Sites, Lemma 00XO. Moreover, j is continuous as well, hence Sites, Lemma 00XR applies.*

Proof. This will follow immediately from the case of algebraic spaces (Properties of Spaces, Lemma 03LP) if we can show that the induced morphism $X_{red} \to Y_{red}$ is étale. Observe that $X \times_Y Y_{red}$ is an algebraic space, étale over the reduced algebraic space Y_{red}, and hence reduced itself (by our definition of reduced algebraic spaces in Properties of Spaces, Section 03E5. Hence $X_{red} = X \times_Y Y_{red}$ as desired. □